

A Finite-Metric Counterexample for Levy–Prokhorov Isometries

Run fa_banach_001, lane 7

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Status

This packet gives a likely valid counterexample to the naive extension of the Banach-space conclusion of Gehér–Titkos [1] to arbitrary complete separable metric spaces. In separable real Banach spaces, every surjective Levy–Prokhorov isometry of $\mathcal{P}_X$ is induced by an affine isometry of X . For a three-point complete separable metric space, we construct a surjective π -isometry of $\mathcal{P}_X$ which sends Dirac masses according to a bijection that is not an isometry of the underlying metric space.

The result should be read with the following scope caveat. The paper’s Banach-space question is answered in the source paper itself. The target here is the later problem prompt asking for characterisations in other complete separable metric settings. The packet does not classify all such spaces; it records a sharp finite obstruction and a complete proof of the resulting counterexample to the Banach-style conclusion.

Source Problem Evidence

The motivating prompt appears in the final remarks of [1], page 18 of the arXiv PDF:

Troyansky theorem [12, 22, 32]. Namely, if $\mu \in \mathcal{F}_X$ and S_μ is contained in the kernel of a strongly exposing functional (for the definition see e.g. [12]), then the equivalence part of Proposition 3.7 can be verified for every element $\hat{x}$ in S_μ . Since by the Lindenstrauss–Troyansky theorem it is easy to see that every $\mu \in \mathcal{F}_X$ can be weakly approximated by such measures, we easily complete the proof of the Main Theorem in this case too. It seems that for finite dimensional spaces, even for the case of 3), we really have to do the whole procedure presented in Section 3, or at least we are not aware of any shortening possibilities.

Finally, we note that throughout Section 3 there were some parts where we considered general complete and separable metric spaces. But later on most of our techniques required that the underlying space had a linear structure. In our opinion it would be interesting to find a characterisation of all surjective π -isometries in the setting of other special (but still general enough) kinds of complete separable metric spaces.

REFERENCES

Transcription

After noting that parts of the proof used general complete separable metric spaces but later techniques required a linear structure, the authors write that it would be interesting to find a characterisation of all surjective π -isometries in other special, still general enough, complete separable metric spaces.

Definitions

For a metric space (X, d) , let $\mathcal{P}_X$ denote the Borel probability measures on X . Gehér–Titkos use Prokhorov’s metric

$$\pi_d(\mu, \nu) = \inf \{ \varepsilon > 0 : \mu(A) \leq \nu(A^\varepsilon) + \varepsilon \text{ for every Borel } A \subseteq X \},$$

where

$$A^\varepsilon = \bigcup_{x \in A} B_\varepsilon(x), \quad B_\varepsilon(x) = \{z \in X : d(x, z) < \varepsilon\}.$$

A map $\Phi : \mathcal{P}_X \rightarrow \mathcal{P}_X$ is a surjective π -isometry if it is onto and $\pi_d(\Phi\mu, \Phi\nu) = \pi_d(\mu, \nu)$ for all $\mu, \nu \in \mathcal{P}_X$.

For a finite set X , total variation will mean

$$\text{TV}(\mu, \nu) = \sup_{A \subseteq X} |\mu(A) - \nu(A)|.$$

Proof Intuition

The Levy–Prokhorov metric only uses open ε -neighborhoods for the scale ε appearing in the inequality. Since every Levy–Prokhorov distance is at most 1, the metric cannot distinguish base-space distances once they have all been pushed past the threshold 1. On a finite metric space whose nonzero distances are all at least 1, every open ball of radius $0 < \varepsilon < 1$ is a singleton. Thus $A^\varepsilon = A$ for every set A , and the Levy–Prokhorov metric collapses to total variation. Total variation is invariant under every permutation of the underlying finite set, including permutations that do not preserve the original metric.

Lemma 1 (Truncation invariance). *Let X be a finite metric space and let $\tau : X \rightarrow X$ be a bijection such that*

$$\tau[A^\varepsilon] = (\tau A)^\varepsilon \quad (0 < \varepsilon < 1, A \subseteq X).$$

Then the push-forward map $\tau_\# : \mathcal{P}_X \rightarrow \mathcal{P}_X$ is a surjective π_d -isometry.

Proof. For every $\varepsilon \geq 1$, the defining Prokhorov inequality is automatic: $\mu(A) \leq 1 \leq \nu(A^\varepsilon) + \varepsilon$. Hence only $0 < \varepsilon < 1$ can affect the infimum below the universal upper bound $\pi_d \leq 1$.

Fix $0 < \varepsilon < 1$. Since τ is a bijection, every subset of X has the form τA for a unique subset A . Using the assumed neighborhood identity,

$$(\tau_\#\mu)(\tau A) = \mu(A) \leq \nu(A^\varepsilon) + \varepsilon = (\tau_\#\nu)((\tau A)^\varepsilon) + \varepsilon.$$

Thus ε is admissible for (μ, ν) if and only if it is admissible for $(\tau_\#\mu, \tau_\#\nu)$. The admissible sets have the same intersection with $(0, 1)$, and both contain every $\varepsilon \geq 1$; therefore their infima are equal. $\square$

Lemma 2 (Large discrete scale). *Let X be a finite metric space such that $d(x, y) \geq 1$ whenever $x \neq y$. Then*

$$\pi_d(\mu, \nu) = \text{TV}(\mu, \nu) \quad (\mu, \nu \in \mathcal{P}_X).$$

Consequently every bijection $\tau : X \rightarrow X$ induces a surjective π_d -isometry $\tau_\#$ of $\mathcal{P}_X$.

Proof. If $0 < \varepsilon < 1$, then $A^\varepsilon = A$ for every subset $A \subseteq X$. Hence such an ε is admissible precisely when

$$\mu(A) - \nu(A) \leq \varepsilon \quad (A \subseteq X).$$

It follows that

$$\pi_d(\mu, \nu) = \sup_{A \subseteq X} (\mu(A) - \nu(A)).$$

The right-hand side equals $\text{TV}(\mu, \nu)$, because taking complements gives $\nu(A) - \mu(A) = \mu(X \setminus A) - \nu(X \setminus A)$. Finally, every bijection of X preserves the collection of subsets and hence preserves total variation. $\square$

Theorem 1 (Concrete counterexample). *There is a complete separable metric space X and a surjective π -isometry $\Phi : \mathcal{P}_X \rightarrow \mathcal{P}_X$ such that Φ is not induced by any isometry of X .*

Proof. Let $X = \{a, b, c\}$ and define

$$d(a, b) = 2, \quad d(a, c) = \frac{5}{2}, \quad d(b, c) = 3,$$

with $d(x, x) = 0$ and d symmetric. The triangle inequalities hold since

$$2 + \frac{5}{2} > 3, \quad 2 + 3 > \frac{5}{2}, \quad \frac{5}{2} + 3 > 2.$$

Thus X is a finite, hence complete and separable, metric space.

Let τ be the transposition swapping a and b and fixing c . This bijection is not an isometry of X , since

$$d(a, c) = \frac{5}{2} \neq 3 = d(\tau a, \tau c).$$

However, all nonzero distances in X are at least 1. By Lemma 2, the push-forward $\Phi = \tau_\#$ is a surjective π_d -isometry of $\mathcal{P}_X$.

It remains only to check that this Φ is not secretly induced by some other isometry of X . If an isometry $\psi : X \rightarrow X$ induced Φ , then for every $x \in X$,

$$\delta_{\psi(x)} = \psi_\# \delta_x = \Phi(\delta_x) = \tau_\# \delta_x = \delta_{\tau(x)}.$$

Dirac masses are injective in their base point, so $\psi(x) = \tau(x)$ for all x . This forces $\psi = \tau$, contradicting the fact that τ is not an isometry. $\square$

Corollary 1. *The conclusion of the Gehér–Titkos Banach-space theorem, with “affine isometry” replaced by “isometry of the underlying metric space”, is false for general complete separable metric spaces.*

Verification Notes

The proof is purely finite and requires no computation. The script `code/check_finite_example.py` checks the triangle inequalities in the displayed three-point metric, verifies that the transposition is not a base-space isometry, and tests total-variation preservation on a small rational grid of probability measures. This script is only a sanity check; the proof above is exact.

Limitations

This is not a complete classification of all complete separable metric spaces whose probability-measure spaces have prescribed π -isometries. It also does not contradict the source paper, whose main Banach-space theorem remains untouched. Rather, it identifies a basic obstruction any broader characterisation must address: the Levy–Prokhorov metric is insensitive to the part of the base metric lying above the truncation scale 1.

A natural next refinement would be to classify finite metric spaces up to the neighborhood data $\{(x, y) : d(x, y) < \varepsilon\}$ for $0 < \varepsilon < 1$, since Lemma 1 shows that this truncated data already creates π -isometries.

Novelty and Literature Check

Before promotion, I searched the run indexes for arXiv:1701.04267, “Levy–Prokhorov”, “Prokhorov metric”, “finite metric space”, and “surjective isometries”. The run indexes showed no prior packet, attempt, or proof-gap entry for this finite-truncation obstruction.

I also ran bounded web searches for combinations of “Levy–Prokhorov finite metric space isometries”, “Prokhorov metric finite metric space total variation”, and “surjective Prokhorov metric isometries metric spaces”. Those searches found the source paper and unrelated uses of total variation on simplices, but no exact prior statement of the counterexample above. Novelty confidence should nevertheless be treated as moderate rather than absolute, because the construction is elementary once the scale-1 truncation is noticed.

Human Review Recommendation

Verify the two scale claims: first, that the Prokhorov defining inequality is automatic for $\varepsilon \geq 1$; second, that if all nonzero base distances are at least 1, then $A^\varepsilon = A$ for $0 < \varepsilon < 1$. After those checks, the reduction to total variation and the non-base-isometry witness are immediate.

References

- [1] G. P. Gehér and T. Titkos, *A characterisation of isometries with respect to the Levy–Prokhorov metric*, arXiv:1701.04267, 2017. <https://arxiv.org/abs/1701.04267>